Music Effects on Mental Health: **Exploring the Impact of Music Metrics on Mental Health Issues**

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Note: This document contains a selected excerpt of the full report. If you would like to read the complete analysis, findings, and supporting visualisations, feel free to get in touch; I'd be happy to share the complete version upon request.

1. Introduction

The connection between mental health and human emotions is a subject that has been increasingly explored as patterns emerge regarding music tastes and people's resultant psychological well-being.

As subscription services such as Spotify, Amazon Music and Deezer continue to dominate the music streaming space, these platforms provide insight into people's listening habits. We aim to analyse whether meaningful correlations can be found between these music streaming preferences and changes in mental health status to uncover whether music can serve as a positive method for emotional well-being.

Our primary question explores: *How does the age, music listening frequency and preferred music genres of people across the globe with specific mental health issues relate to music preferences and mood improvement?*

The following paper presents our methodology, data visualisations and entity relationship diagram to fully understand whether a causal connection can be derived. We can then leverage this information for future guidance for these streaming services, regarding what attracts people to their chosen platform. In particular, our insights will assist with what genres are especially important in improving the wellbeing of specific age groups.

2. Dataset Selection and Rationale

The datasets were all obtained from two Kaggle sources named *Music & Mental Health Survey Results* and *Global Music Streaming Trends & Listener Insights*.

The streaming dataset covers listening behaviors and preferences from 2018–2024 across major platforms (Spotify, Apple Music, Tidal). In terms of the survey, collected by questionnaires, the *Music & Mental Health Survey Results* dataset includes the frequency of listening to 16 types of music, as well as four mental health scores for certain mental health issues (anxiety, depression, insomnia, and OCD).

The addition of the *Music & Mental Health Survey Results* dataset offers a multi-level view, allowing insights into how genre preference and time-of-day listening relate to mental-health outcomes. Both datasets were connected to explore the association between music listening behavior, streaming patterns, and mental health issues. Together, the combined dataset design brings multiple views together, helping us to understand and explain when, how, and why people listen to music, and how music influences mental well-being.

Figure 1 — Music & Mental Health — Data Overview Table

Age	Primary streaming service	Hours per day	Fav genre	Exploratory	BPM	Music effects	Instrumentalist	Composer
group people by age and see links with music habits or mood.	Respondents' primary streaming platform used, comparing their cross platform behavior	The number of hours the respondent listens to music per day, which shows daily listening time and its relationship with emotions or study/work	Favorite or top music genre,which explain music taste and how different styles may influence emotions	If the respondents are actively explore new artists/genres? This may be related to emotions or creativity.	Preferred tempo (BPM).Used to study whether the beats of music affect emotions.	records how music helps them , allowing us to compare the impact of music on different mental issues.	If the respondent play an instrument regularly have an impact on feelings.	The respondent regularly composes music, which helps us explore how creative activity may relate to emotional expression.

Data Overview Table

Created using data provided by *Music & Mental Health Survey Results*.

Figure 2 — Global Music Streaming — Data Overview Table

User_ID	Age	Country	Streaming Platform	Top Genre
Unique identifier for each listener,allow us to link the same person across tables.	Age of the listener,which allowing us grouped by age and compare habits or preferences across ages.	Country where the user resides,enables regional comparisons and showing country-level differences.	The platform used,which helps compare behavior across platforms	The genre the listener streams the most.Shows taste patterns
Minutes Streamed Per Day	**Number of Songs Liked**	**Most Played Artist**	**Subscription Type**	**Listening Time (Morning/Afternoon/Night)**
Average daily listening time	Total liked songs by the user,make genre groups.	Artist the listener plays the most.Identifies different preferences and distinguishes niche vs. mainstream tastes.	Free or Premium subscription.Helps compare listening habits between free and paid users.	Time of day they listen the most . Helps us to understand daily routine patterns and find the association between specific time and emotions.

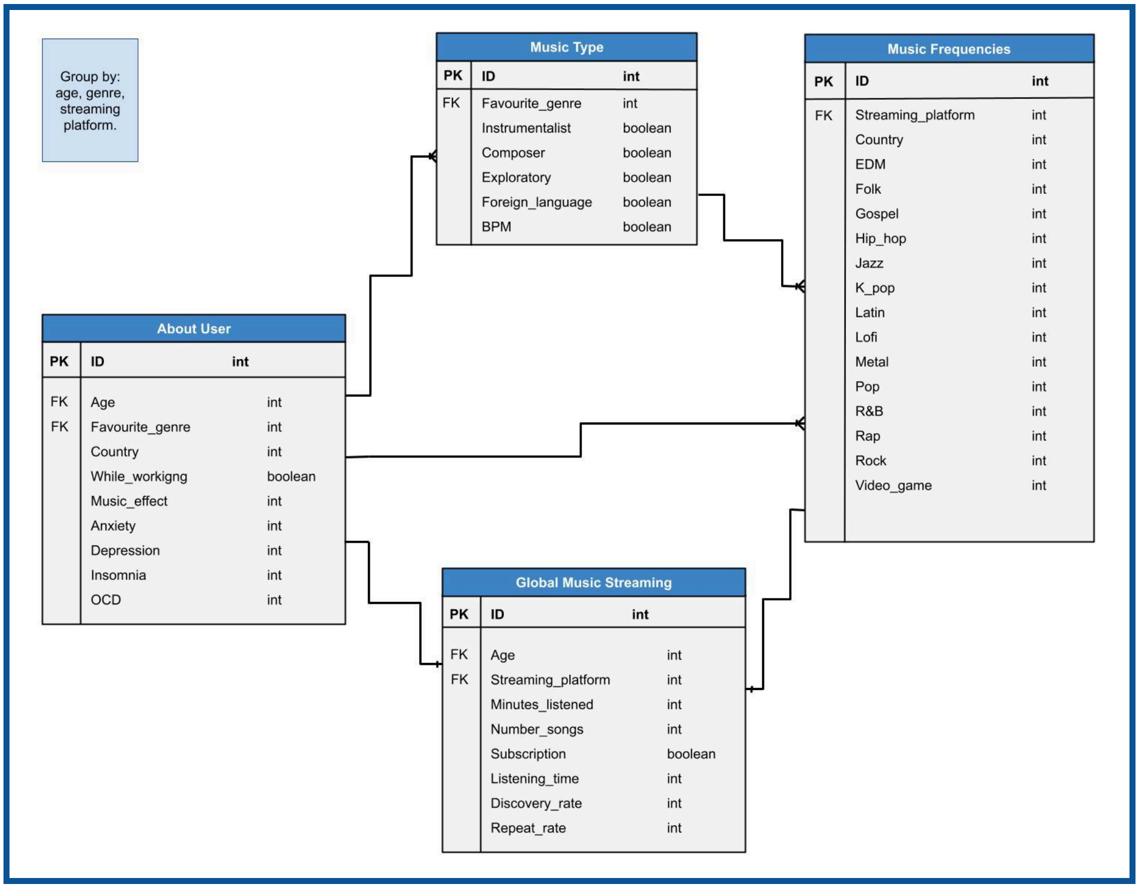
Data Overview Table

Created using data provided by *Global Music Streaming Trends & Listener Insights*.

3. Relationships Between Entities

The ERD shown in Figure 3 consists of four main entities, categorised by the user ID as their Primary Key (PK) and their corresponding attributes. The Logical Database Model (LDM) can be found in the Appendix as Figure 5 of the full report.

Figure 3 - Entity-Relationship Diagram

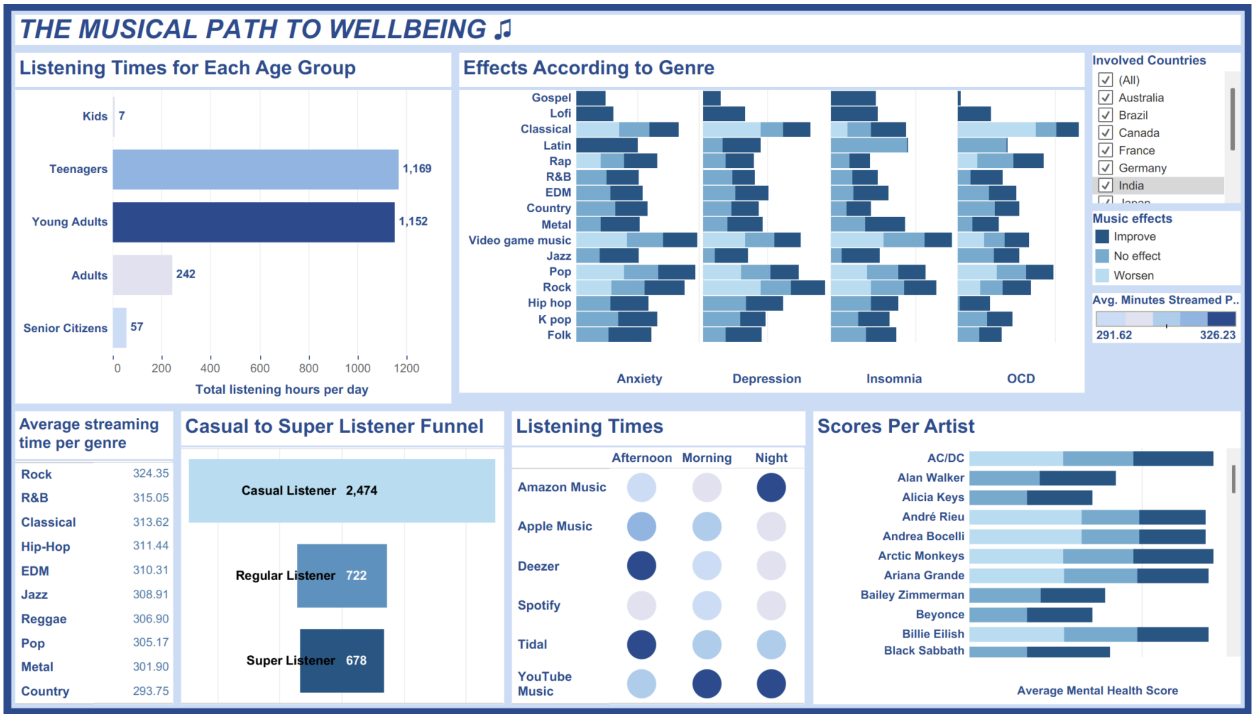


ERD

Created using data provided by Global Music Streaming Trends & Listener Insights on Google Drawings.

4. Visualisation and Graphic Ideas

Figure 4 - Tableau Dashboard



The overall intent of our dashboard in Figure 4, is to deliver the content in a clean yet effective way using simple visualisations with thoughtful additions. For the dashboard, we chose a card based layout over solid colour: the colours were chosen to create cohesion while also being colourblind friendly. Blue is often associated with calmness, trust and wellbeing, which aligns with our overall theme of mental wellbeing. The white spaces prevent cognitive overload and create clear visual grouping, which is also assisted by the margins created by the background which flows throughout the dashboard.

We chose the *Arial* as our primary font for the whole dashboard, as *Arial* is a Sans Serif font, meaning that it lacks decorative strokes in its design, thus providing high legibility, while also not being distracting. It is a widely used font which makes it compatible with multiple platforms. This, combined with the font being evenly sized and used properly in bold or italics, makes for a pleasant reading experience for the end user.

The dashboard is connected by the field *Age Group* in the *Listening Times for Each Age Group* though a common filter, such that when clicking on any of the age groups, the entire dashboard will respond with information pertaining to that particular age group. This is accompanied by a filter which can be used to look into the data with a specific country in mind. We have also added tooltips in all of the visualisations to either provide more data or context to the user in case the data cannot be explained just by using the axis.

5. Condensed Data Results

Age

Teenagers and adolescents showed the highest music listening duration, while mental health issues generally decreased with age. This revealed a strong relationship between listening habits, age, and mental health outcomes, making younger age groups the most significant focus for further analysis.

Frequency

Individuals with higher mental health issue scores also tended to engage in longer listening sessions. Participants who reported positive effects from music demonstrated the highest average listening duration, suggesting that immersive listening habits may support emotional regulation and stress relief.

Music Genres

The impact of music varied significantly across genres. Gospel and Lofi music showed positive effects across all four mental health categories analysed, while Pop and Video Game music showed the most negative outcomes. Classical music did not consistently produce positive effects and was associated with negative outcomes for OCD. Latin music showed benefits for anxiety and depression, but not for insomnia or OCD. Among younger listeners, Hip-hop, Folk, and Jazz demonstrated particularly strong therapeutic effects for teenagers, while Classical music appeared more beneficial for young adults. Lastly, gospel music was highly rated among teenagers but showed little appeal for young adults.

Listener and Membership Type

The gap between casual listeners and “super listeners” was relatively small, suggesting that many casual users could be converted into highly engaged listeners through more targeted music recommendations and personalised streaming experiences.

6. Recommendations and Conclusions

The analysis identified clear relationships between music listening habits and mental health outcomes, with age, listening behaviour, and genre preference emerging as key influencing factors. The findings suggest that music may play a meaningful role in emotional wellbeing, although the effects vary considerably across listener groups and genres.

Several patterns highlighted the importance of personalised listening experiences, particularly among younger audiences, where engagement levels and emotional responses to music appeared strongest. The results also suggest opportunities for streaming platforms to better tailor recommendations and wellbeing-focused experiences based on listener behaviour and preferences. In addition, listener engagement trends indicated strong potential for converting casual users into more active listeners through targeted music discovery and genre personalisation strategies.

Further insights, detailed analysis, and recommendations are explored in the full report.